Hung Q. Ngo

Curriculumn Vitae

RelationalAI
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Education

1996–2001 **Ph.D. in Computer Science & Engineering**, *University of Minnesota*, Twin Cities
Dissertation: "Issues in Interconnection Networks", Advisor: Ding-Zhu Du

1999–2001 **M.S. in Mathematics**, *University of Minnesota*, Twin Cities
Thesis: "**P**-Species and the q-Mehloer Formula", Advisor: Dennis Stanton

1990–1995 **B.S. in Computer Engineering**, *Ho Chi Minh city University of Technology*, Ho Chi Minh city, Vietnam

Employment History

2017–present **VP Research, led the design and implementation of query optimizer**, *RelationalAI Inc.*

2015–2017 **Computer Scientist**, *LogicBlox Inc.*, Runtime Group

2001–2017 **Assistant→Associate→Full Professor**, *Computer Science and Engineering, SUNY Buffalo*

Awards and Honors

2025 **ACM Distinguished Member**

2025 **Alberto Mendelzon Test of Time Award**, *(For the 2015 "Joins via Geometric Resolutions: Worst-case and Beyond" paper)*

2022 **Alberto Mendelzon Test of Time Award**, *(For the 2012 "Worst-case Optimal Join" paper)*

2022 **Best paper award**, *PODS 2022*

2022 **ACM SIGMOD Research Highlights Award**, *(For the Datalogo paper)*

2022 **Best paper award**, *ICDT 2019*

2018 **Gems of PODS**, *Invited to give the "Gems of PODS" talk on Worst-Case Optimal Join Algorithms*

2016 **ACM SIGMOD Research Highlights Award**, *(For the Functional Aggregate Queries paper)*

2016 **Best paper award**, *PODS 2016*

2016 **CSE Sustained Teaching Execellence Award**

2012 **Best paper award**, *PODS 2012*

2008 **Best paper award**, *COCOON 2008*

2005 **Exceptional Scholar (Young Investigator) Award**, *SUNY Buffalo*

2004–2009 **NSF CAREER Award**

2002–2003 **Outstanding Teacher Awards**, *Fall 2002, Spring 2003, Fall 2003*
By the Graduate Student Association of the Department of Computer Science & Engineering, SUNY Buffalo

2000–2001 **Guidant Fellowship for outstanding achievement**
One fellowship was awarded by the Department of Computer Science and Engineering, University of Minnesota

1990 **First prize**, *Ho Chi Minh city Mathematical Olympiad*

1989–1990 **Various prizes**, *Vietnamese Mathematical Olympiads*

1988 **Second prize**, *Ho Chi Minh city Physics Olympiad*, (No first prize awarded)

Selected Professional Activities

Simons Program (2023)	**Co-Organizer** Logic and Algorithms in Database Theory and AI
PODS 2023	**PC Chair** The 41st ACM Symposium on Principles of Database SystemsTheory
ICDT 2022	**TPC member** The 25th International Conference on Database Theory
VLDB 2021	**TPC member** International Conference on Very Large Data Bases
PODS 2020	**TPC member** The 38th ACM Symposium on Principles of Database SystemsTheory
PODS 2019	**TPC member** The 37th ACM Symposium on Principles of Database SystemsTheory
PODS 2018	**TPC member** The 36th ACM Symposium on Principles of Database SystemsTheory
ICDT 2017	**TPC member** International Conference on Database Theory
2010–2015	**Associate Editor** Discrete Mathematics, Algorithms, and Applications
INFOCOM 2008–2015	**TPC member** IEEE Conference on Computer Communications (INFOCOM)
COCOON 2009	**Conference and program chair** The 15th International Conference on Combinatorics and Computing (COCOON)
WoWMoM 2006	**Local arrangement chair** IEEE International Symposium on a World of Wireless, Mobile and Multimedia Networks
2011–2014	**Co-organizer** Coding, Complexity, and Sparsity Workshop
GLOBECOM 2011	**TPC member** IEEE Global Communications Conference
ICC 2008–2010	**TPC member** IEEE International Conference on Communications
ISAAC 2009	**TPC member** International Symposium on Algorithm and Computation
AINA 2007–2009	**TPC member** IEEE International Conference on Advanced Information Networking and Applications
HPSR 2006	**TPC member** IEEE Workshop on High Performance Switching and Routing
COCOON 2003	**TPC member** International Conference on Combinatorics and Computing
NSF Panels	**Panelist** Several NSF panels

Selected Departmental Services

2012–2015	**Director of Graduate Studies**
2012–2015	**CSE Executive Commitee**
2012–2015	**Alternate Member of SEAS Tenture Committee**
2001–2006 2007–2012	**Graduate Affairs Committee**

2011–2012 **Faculty Search Committee**
2002–2006

2006–2007 **Undergraduate Affairs Committee**

2002–2003 **Library Commitee**
2008–2010

2003–2004 **Internship Commitee**

2004–2005 **Chair of the Colloquium Committee**

2001–2002 **Facilities Committee**

Research Supervision

2013–2016 **Mahmoud Abo Khamis**, *Ph.D. student*, First employer: LogicBlox
Thesis title: "FAQ: Querstions Asked Frequently"

2006–2011 **Thanh-Nhan Nguyen**, *Ph.D. student*, First employer: IBM
Thesis title: "Algorithms for some Network Design Problems"

2006–2011 **Anh N. Le**, *Ph.D. student*, First employer: a startup
Thesis title: "On Data Flow Masquerading"

2005–2010 **Yang Wang**, *Ph.D. student*, First employer: Google
Thesis title: "New Results in the Design and Analysis of Non-blocking Switching Networks"

2004–2009 **Duc T. Ha**, *Ph.D. student*, First employer: Sumo Logic
Thesis title: "Propagating Malicious Codes: Theory and Experiments"

2001–2006 **Dazhen Pan**, *Ph.D. student*, First employer: Microsoft
Thesis title: "Complexity and Constructions of WDM Switching Networks"

2003–2005 **Tuong Nguyen**, *M.S. student*, Currently with M_Service JSC, Vietnam

2003–2005 **Harleen Dhillon**, *M.S. student*, Currently Vice President at Ennovance Capital LLC
Thesis title: "On reducing control overhead in on-demand multicast routing protocol"

2001–2003 **Vikas P. Verma**, *M.S. student*, With FCS Business Solulutions
Thesis title: "Distributed Algorithms for Computing Connected Dominating Sets."

2001–2003 **Purnima M. Mavinkurve**, *M.S. student*, First employer: Amazon
Thesis title: "Centralized and Distributed Algorithms for Power-Conserving Multicasting in Static Wireless Ad Hoc Networks"

Ph.D. Dissertation Committee Member

2004 **Xiaojun Cao**, *Advisor: Chunming Qiao*, Defended

2005 **Huaming Zhang**, *Advisor: Xin He*, Defended

2005 **Ramkumar Chinchani**, *Advisor: Shambhu Upadhyaya*, Defended
Sumesh Philip, *Advisor: Chunming Qiao*, defended May 2005
Xiang Yu, *Advisor: Chunming Qiao*, Defended

2006 **Guang Xu**, *Advisor: Jinhui Xu*, Defended

2007 **Zhenming Chen**, *Advisor: Jinhui Xu*, Defended
Peng Lin, *Advisor: Chunming Qiao*, Defended

2008 **S. Vidyaraman**, *Advisor: Shambhu Upadhyaya*, Defended
Yulai Xie, *Advisor: Jinhui Xu*, Defended
Mingen Lin, *Advisor: Jinhui Xu*, Defended

2009 **Xuming Lu**, *Advisor: Murat Demirbas*, Defended
Madhusudhanan Chandrasekaran, *Advisor: Shambhu Upadhyaya*, Defended
Xin Liu, *Advisor: Chunming Qiao*, Defended
Seokhooon Yoon, *Advisor: Chunming Qiao*, Defended
Xi Zhang, *Advisor: Jan Chomicki*, Defended

Sunu Mathew, *Advisor: Shambhu Upadhyaya*, Defended

2010 **Asheq Khan**, *Advisor: Satish Tripathy*, Defended

Onur Soyal, *Advisor: Murat Demirbas*, Defended

2011 **Steve Uurtamo**, *Advisor: Atri Rudra*, Defended

Yongding Zhu, *Advisor: Jinhui Xu*, Defended

2012 **Jia Zhao**, *Advisor: Chunming Qiao*, Defended

Lei Xu, *Advisor: Jinhui Xu*, Defended

2013 **Albert Chen**, *Advisor: Jason Corso*, Defended

Liang Ge, *Advisor: Aidong Zhang*, Defended

Swapnooneel Roy, *Advisor: Atri Rudra*, Defended

2014 **Manavender Reddy**, *Advisor: Venu Govindaraju*, Defended

Caiming Xiong, *Advisor: Jason Corso*, Defended

Aditya Wagh, *Advisor: Chunming Qiao*, Defended

Utkarsh Porwal, *Advisor: Venu Govindaraju*, Defended

Dung Nguyen, *Advisor: Alan Selman*, Defended

Hu Ding, *Advisor: Jinhui Xu*, Defended

2015 **Zilong Ye**, *Advisor: Chunming Qiao*, Defended

Arti Shivram, *Advisor: Venu Govindaraju*, Defended

Yingbo Zhou, *Advisor: Venu Govindaraju*, Defended

2016 **Jiun-Jie Wang**, *Advisor: Roger He*, Defended

Vishrawas Gopalakrishnan, *Advisor: Aidong Zhang*, Defended

Devansh Arpit, *Advisor: Venu Govindaraju*, Defended

Qi Li, *Advisor: Jing Gao*, Defended

M.S. Thesis Committee Member

2003 **Ranjani Sridharan**, *Advisor: Ramalingam Sridhar*, Defended

2004 **Aarthie Muthukrishnan**, *Advisor: Shambhu Upadhyaya*, Defended

2005 **Aruna Balasubramanian**, *Advisor: Ramalingam Sridhar*, Defended

2005 **Madhusudhanan Chandrasekaran**, *Advisor: Shambhu Upadhyaya*, Defended

Grant Support

2014–2018 **National Science Foundation (NSF)**, *Proposal # CNF-1409551*, **PI**, $1,215,973
"TWC: **Medium**: Collaborative: Data is Social: Exploiting Data Relationships to Detect Insider Attacks." (UB co-PIs: Varun Chandola, Oliver Kennedy, Shambhu Uphadyaya. Michigan co-PI: XuanLong Nguyen.) UB's share $975,999.00. 2014–2018

2013–2016 **National Science Foundation (NSF)**, *Proposal # CCF-1319402*, **PI**, $499,999
"AF:III:Small:Collaborative Research: New Frontiers in Join Algorithms: Optimality, Noise, and Richer Languages." (UB Co-PI: Atri Rudra, Stanford Co-PI: Chris Ré.) UB's share $326,101.00.

2012–2015 **National Science Foundation (NSF)**, *Proposal # CCF-1161196*, **co-PI**, $1,199,230
"AF: **Medium**: Collaborative Research: Sparse Approximation: Theory and Extensions." (Lead PI: Anna Gilbert, Michigan; UB's co-PI: Atri Rudra, Rutger's co-PI: Muthu Muthukrishnan). UB's share $305,467.

2004–2009 **National Science Foundation (NSF)**, *Proposal # CCF-0347565*, **PI**, $409,999
"CAREER: Designs and Analyses of WDM Switching Architectures."

2004–2006 **The Defense Advanced Research Projects Agency (DARPA)**, **co-PI**, $1,292,295
"Mitigating the Insider Threat using High-dimensional Search and Modeling." (Lead PI: Eric Van Den Berg, Telcordia Technologies; UB's PI: Shambhu Upadhyaya. We were a subcontractor from Telcordia. The other subcontractor is R. Maxion, CMU.) UB's share $ 255,862.

2012–2013 **Gift from LogicBlox Inc.**, $35,000

Selected Invited Talks

AAIT 2022 **Online Seminar on Algorithmic Aspects of Information Theory**, *On an Information Theoretic Approach to Cardinality Estimation*

UC Berkeley 2022 **Sky Computing DB Seminar**, *On an Information Theoretic Approach to Cardinality Estimation*

ICDT 2022 **Keynote**, *On an Information Theoretic Approach to Cardinality Estimation*

NorthEastern Univ 2020 **Database Seminar**, *Answering (Functional Aggregate) Queries via Tensor Decomposition*

PODS 2018 **Gems of PODS**, *Worst-case Optimal Join Algorithms: Techniques, Results, and Open Problems*
The 37th Symposium on Principles of Database Systems. Houston, TX, U.S.A.

AMW 2018 **Keynote**, *Worst-case Optimal Join Algorithms: Techniques, Results, and Open Problems*
The 12th Alberto Mendelzon International Workshop on Foundations of Data Management, Cali, Colombia, May 22–25, 2018.

Oxford 2017 **Information Systems Seminar**, *Disjunctive Datalog, Shannon-type Inequalities, and Submodular Width*
Department of Computer Science, Oxford University, UK. Fall 2017.

HIGHLIGHTS 2017 **Keynote**, *Disjunctive Datalog, Shannon-type Inequalities, and Submodular Width*
Highlights of Logic, Games, and Automata, Sep 2017, London, U.K.

Stanford 2017 **Logic Group Seminar**, *Functional Aggregate Queries are FAQs*
Computer Science Department, Stanford University, Spring 2017.

Berkeley 2017 **RISE Lab Seminar**, *Way Beyond Worst-Case Optimal Algorithms*
Computer Science and Engineering, UC Berkeley, Spring 2017.

UCSD 2016 **Database Seminar**, *Functional Aggregate Queries Asked Frequently*
Department of Computer Science, UCSD, Spring 2016.

Wisconsin 2016 **Database Seminar**, *Functional Aggregate Queries Asked Frequently*
Department of Computer Science, University of Wisconsin at Madison, Spring 2016.

Oxford 2015 **Database and Knowledge Representation Seminar**, *Functional Aggregate Queries Asked Frequently*
Department of Computer Science, Oxford University, UK. Fall 2015.